

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>

1. Convert the kinetic energy of the electron from eV to Joules: $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

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Kinetic energy = $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

2. Use the kinetic energy to find the momentum of the electron :

Kinetic energy = $(\text{momentum}^2) / (2 \times \text{mass})$

momentum = $\sqrt{2 \times \text{mass} \times \text{kinetic energy}}$

mass of an electron = $9.11 \times 10^{-31} \text{ kg}$

momentum = $\sqrt{2 \times 9.11 \times 10^{-31} \text{ kg} \times 1.6 \times 10^{-19} \text{ J}}$

momentum = $1.01 \times 10^{-24} \text{ kg m/s}$

3. Use the momentum to find the de Broglie wavelength :

de Broglie wavelength = $h / \text{momentum}$

$h = 6.626 \times 10^{-34} \text{ J s}$

de Broglie wavelength = $6.626 \times 10^{-34} \text{ J s} / 1.01 \times 10^{-24} \text{ kg m/s}$

de Broglie wavelength = $6.56 \times 10^{-10} \text{ m}$

Convert the de Broglie wavelength from meters to centimeters :

de Broglie wavelength = $6.56 \times 10^{-10} \text{ m} \times 100 \text{ cm/m}$

de Broglie wavelength = $6.56 \times 10^{-8} \text{ cm}$

</think><answer> A. 2.48×10^{-8} cm

